Lab Report

Only for course Teacher						
		Needs Improvement	Developing	Sufficient	Above Average	Total Mark
Allocate mark & Percentage		25%	50%	75%	100%	25
Problem understanding & Analysis	5					
Implementation	10					
Report writing	10					
					Total obtained mark	
Comments						

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Student Name: Md. Tuhid Hasan Sakib
Student ID: 232-35-597
Section: 41_G1
Course Code: SE 222
Course Title: Software Engineering Web Application Lab

Course Teacher Name: Pranto Saha
Designation: lecturer
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Software Requirements Specification

for

Simple Blog & Article Platform

Version 1.0

Prepared by 232-35-597

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1. Introduction

1.1 Purpose

The purpose of this document is to provide a comprehensive overview of the Simple Blog & Article Platform Project. It is a multi-user blogging platform designed to provide a seamless experience for administrators, authors, and visitors alike. The system focuses on content management, role-based access control, and interactive communication through comments.

1.2 Objective

- To develop a robust multi-user blog system using Laravel 10.
- To implement a Role-Based Access Control (RBAC) system (Admin, Author, Visitor).
- To provide advanced features like soft deletes, media management, and nested comments.
- To ensure a responsive and dynamic user interface using the Oredoo template.

1.3 References/Related Works

This project utilizes the Laravel 10 framework, PHP 8.1+, and the Eloquent ORM. It follows the MVC (Model-View-Controller) architecture to ensure scalability and maintainability. The frontend is powered by the Oredoo professional blog template.

2. Overall Description

2.1 Product Perspective

Simple Blog & Article Platform is a web-based application built on the Laravel framework. It functions as a standalone content management system where users can register, write articles, and interact with others. The system is managed through a secure dashboard.

2.2 Motivation

The primary motivation is to create a modern, high-performance blogging engine that simplifies the process of content publishing while maintaining high security and data integrity through features like password hashing and CSRF protection.

2.3 User Classes and Characteristics

1. **Admin:** Has full control over users, categories, tags, settings, and all content.
2. **Author:** Can create, manage, and edit their own posts through a personalized dashboard.
3. **Visitor:** Can read posts, search content, and engage via the commenting system.

2.4 Operating Environment

- **Server:** PHP 8.1 or higher.
- **Database:** MySQL (Eloquent ORM used).
- **Framework:** Laravel 10.
- **Browser:** Any modern web browser (Chrome, Firefox, Safari).

3. Project Features and Outcomes

3.1 List of Features

- Multi-Role System (Admin, Author, Visitor)
- Advanced Post Management with featured posts and view counters
- Comment System with nested replies and guest support
- Media Library for centralized image uploads
- Soft Deletes for Posts, Categories, and Comments
- Dynamic Menu and Site Settings management

3.21 Database Models

The system consists of 10 primary models:

- **User:** Handles authentication and roles.
- **Post:** Manages article content and metadata.
- **Category & Tag:** For content organization.
- **Comment:** Supports threaded discussions.
- **Media:** Tracks uploaded files.
- **Site Setting & social media:** For dynamic configuration.

4. External Interface Requirements

4.1 User Interfaces

The interface is divided into two parts:

- **Frontend:** Public-facing blog with search, category archives, and post views.
- **Dashboard:** A secure area for Admin and Authors to manage resources.

4.2 Input Output Demo

Inputs include post data (title, slug, content, thumbnail), user registration details, and comments. Outputs include the rendered blog pages, view counts, and calculated reading times.

5. Future Scope and Limitation

5.1 Limitation

- No native mobile application.
- API implementation is currently limited (routes exist but functionality is minimal).
- Limited to 10 featured posts on the homepage.

5.2 Future Scope

- Implementing a full RESTful API for mobile support.
- Adding an email notification system for new comments.
- Integration of an advanced SEO optimization tool within the post editor.

5.3 Conclusion

The Simple Blog & Article Platform Project is a well-structured, feature-rich platform. It successfully implements essential web engineering practices including RBAC, MVC, and data integrity measures, making it a professional-grade blogging solution.

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